

1. Suppose

$$a \equiv a' \pmod{n}, \quad b \equiv b' \pmod{n}.$$

Prove that

$$a + b \equiv a' + b' \pmod{n}$$

and

$$ab \equiv a'b' \pmod{n}.$$

Conclude that

$$[a] + [b] = [a + b], \quad [a][b] = [ab]$$

are well-defined operations on  $\mathbb{Z}/n\mathbb{Z}$ .

2. Consider

$$\mathbb{Z}/8\mathbb{Z}.$$

For each element

$$[0], [1], \dots, [7],$$

determine whether it has a multiplicative inverse. For every invertible element, explicitly find its inverse. Then formulate a conjecture relating invertibility of  $[a]$  to

$$\gcd(a, 8).$$

Do not prove the general conjecture yet.

3. Let  $n > 1$  be composite. Prove directly that there exist

$$[a] \neq [0], \quad [b] \neq [0]$$

in  $\mathbb{Z}/n\mathbb{Z}$  such that

$$[a][b] = [0].$$

Then prove that this implies  $\mathbb{Z}/n\mathbb{Z}$  cannot be a field. Try to write this as one clean implication chain rather than relying on an example.

4. Let  $p$  be prime and let

$$[a] \in \mathbb{Z}/p\mathbb{Z}, \quad [a] \neq [0].$$

Using Bézout's identity,

$$ax + py = \gcd(a, p)$$

for suitable  $x, y \in \mathbb{Z}$ , prove that  $[a]$  has a multiplicative inverse. Conclude:

$$\boxed{\mathbb{Z}/n\mathbb{Z} \text{ is a field} \iff n \text{ is prime.}}$$

This is today's main proof.

5. Work in  $\mathbb{F}_5$ . Compute

$$1^4, \quad 2^4, \quad 3^4, \quad 4^4$$

modulo 5. Then formulate a conjecture for

$$a^{p-1} \pmod{p}$$

when  $p$  is prime and

$$pa.$$

Do not try to prove the general statement yet. This observation will lead to Fermat's little theorem, finite multiplicative groups, and later the Frobenius map

$$x \mapsto x^p,$$

which becomes a central actor in the function-field version of RH.